Contact

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Top Skills

Market Opportunities
Innovation Management
Aerospace

Patents

Bond lead
Method and apparatus for manufacturing a wind turbine blade component with uniform temperature curing

Andrew Limmack

Chief Engineer - Aerospace Propulsion & Advanced Systems
Loxton, England, United Kingdom

Summary

Aerospace Engineer with a passion for innovation in advanced materials technology, harsh environment testing and protection and smart/multifunctional materials and systems.

Extensive experience in nano materials, development and optimisation of high efficiency manufacturing methods for medium to very large composite structures, engineering and technology feasibility studies, human-system interaction and sustainable and environmentally friendly technologies.

Currently particularly active in civil aerospace, defence, renewable energy (wind) and automotive fields but particularly keen to conduct further work in medical, space and electric transport sectors.

Experience

National Composites Centre
7 years 11 months

Chief Engineer - Aerospace Propulsion
November 2021 - Present (4 years 3 months)
Bristol, England, United Kingdom

Deputy Chief Engineer - Aerospace & Defence (Defence & Propulsion)
July 2019 - November 2021 (2 years 5 months)
Bristol, United Kingdom

Engineering Capability Lead (interim)
March 2019 - July 2019 (5 months)
Bristol, United Kingdom

ECL for Material Systems team

Principal Research Engineer
March 2018 - March 2019 (1 year 1 month)
Bristol, United Kingdom

Limmack Ltd.

Consultant Engineer

August 2016 - Present (9 years 6 months)

Bristol, United Kingdom

Advanced materials: feasibility studies and exploitation routes, laboratory advice, equipment and process design, materials development, smart and multifunctional materials predominately based on polymer systems with embedded nano and micro scale technology.

Traditional composites: manufacturing optimisation, automated manufacturing, novel bonding, specialised testing.

Harsh environment materials (protection) and testing. Design of harsh environment testing equipment. including erosion, impact and aircraft fire performance.

Project services: technical specification writing, funding application writing, procurement, project management, strategy and exploitation.

Sustainability: lower energy processing, substitution of harmful materials, reduction of waste,

University of Bristol

Research Collaborator - Composites & Nanotechnology

March 2011 - November 2021 (10 years 9 months)

Provide guidance and assistance for research activities in nanotechnology and advanced composites.
Teach nanoparticle functionalisation, dispersion and nanocomposite processing.
Undertake formulation work to support industrial contracts in nanotechnology.
Advise on procurement of nanocomposite processing equipment.

National Composites Centre

Consultant Research Engineer (Limmack Ltd)

August 2016 - January 2018 (1 year 6 months)

Consultant research engineer proudly assisting the National Composites Centre with specialist knowledge on cutting edge projects, feasibility studies and expansion of automated manufacturing and inspection capabilities.

Self Employed

Nanotech, advanced materials and systems consultant
April 2011 - August 2016 (5 years 5 months)

Advanced materials: feasibility studies and exploitation routes, laboratory advice, equipment and process design, materials development, smart and multifunctional materials predominately based on polymer systems with embedded nano and micro scale technology.

Traditional composites: manufacturing optimisation, novel bonding, specialised testing.

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Project services: funding application writing, management, strategy and exploitation.

Sustainability: lower energy processing, substitution of harmful materials, reduction of waste,

LMAT Ltd
Senior Engineer - Materials Design & Development
March 2011 - October 2015 (4 years 8 months)

Perform R&D into smart and multifunctional nanocomposite materials.
Development of coatings for harsh environments.
Support composite manufacturing process simulation through characterisation of composite components.
Development of novel sensor systems for investigation of composite processing defects.
Thermal analysis of existing and development of new composite curing methods.

University of Bristol
1 year 4 months

Research Assistant
January 2009 - February 2010 (1 year 2 months)

Development of new technology high resistance bonding leads for composite Aircraft. Investigation of current materials and solutions that could be used or adapted and invention of new novel nanotechnology route.

Materials Scientist

November 2008 - March 2009 (5 months)

Characterisation and ageing analysis of phenolic resin and filled phenolic formulations through rheometry.

Education

University of Bristol

PhD, Nanocomposite materials, Aerospace engineering · (2004 - 2010)

University of Bristol

M.Eng, Aerospace Engineering · (1999 - 2004)